

CliniScan: Chest X-ray Lung Abnormality Detection System

Milestone 3 Report: Model Training, Evaluation, and Analysis
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Table of Contents

1. Introduction
 2. Model Selection and Training
 3. Training Observations
 4. Performance Evaluation
 5. Prediction and Visualization
 6. Error Analysis
 7. Key Insights
 8. Conclusion
 9. Future Work
 10. Summary
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1. Introduction

This milestone focuses on the implementation, training, and evaluation of object detection models using the YOLOv8 architecture for the CliniScan medical imaging pipeline. The objective was to compare two model variants:

- **YOLOv8x (Extra Large)** — the most powerful YOLOv8 variant
- **YOLOv8s (Small)** — a lightweight yet capable variant

The goal was to analyze their effectiveness on the VinBigData Chest X-ray dataset containing 14 disease classes, understand the trade-offs between model complexity, training time, and detection accuracy, and identify the most practical model for hardware-constrained environments.

The work builds upon the preprocessing pipeline developed in earlier milestones and aims to generate meaningful insights from model predictions on real medical images.

2. Model Selection and Training

Two variants of the YOLOv8 model were used for comprehensive comparative analysis:

- **YOLOv8x (Extra Large)** — Deepest architecture with the highest parameter count, designed for maximum accuracy at the cost of compute.
- **YOLOv8s (Small)** — Efficient architecture balancing speed and detection quality, suitable for hardware-limited environments.

Training Configuration

Parameter	YOLOv8x	YOLOv8s
Epochs	25	25
Image Size	512×512	512×512
Batch Size	4	2
Optimizer	AdamW (Auto)	AdamW (Auto)
Device	NVIDIA GPU (device=0)	NVIDIA GPU (device=0)
Workers	8	0
Mixed Precision (AMP)	Enabled	Default
Pretrained Weights	COCO (YOLOv8x.pt)	COCO (YOLOv8s.pt)

Dataset

Split	Images
Training	~4,000
Validation	~1,000
Test	260
Total Classes	14

Classes: Aortic Enlargement, Atelectasis, Calcification, Cardiomegaly, Consolidation, ILD, Infiltration, Lung Opacity, Nodule/Mass, Other Lesion, Pleural Effusion, Pleural Thickening, Pneumothorax, Pulmonary Fibrosis.

3. Training Observations

YOLOv8x Observations

- Box loss started high (~2.77) and gradually decreased across epochs.
- Class loss showed instability in mid-epochs due to the raw complexity of 14-class medical detection.
- DFL loss declined steadily demonstrating geometric learning of bounding box distributions.
- mAP@50 remained very low (~0.007) during the early epochs of the 14-class run, indicating the model needed more epochs to converge on such a complex multi-class task.

YOLOv8s Observations

- Box loss reduced consistently from **2.23** $\rightarrow$ **1.62** across 14 epochs.
- Class loss decreased from **3.48** $\rightarrow$ **1.89**, showing strong class feature learning.
- DFL loss fell from **1.97** $\rightarrow$ **1.52**, indicating efficient localization learning.
- mAP@50 showed a clear positive trend: **0.087** $\rightarrow$ **0.200** across epochs.
- Training speed was approximately **~2h 20min** for 14–25 epochs on RTX 3050 4GB VRAM.

4. Performance Evaluation

4.1 Final Metrics — YOLOv8s (Validation)

Metric	Value
Precision	0.304
Recall	0.275
mAP@50	0.241
mAP@50-95	0.121

4.2 Per-Class Performance — YOLOv8s

Class	Precision	Recall	mAP@50	mAP@50-95
Aortic Enlargement	0.706	0.569	0.670	0.403
Cardiomegaly	0.739	0.560	0.672	0.447
Pleural Effusion	0.277	0.445	0.280	0.131
Atelectasis	0.169	0.312	0.339	0.131
ILD	0.203	0.396	0.233	0.110
Infiltration	0.439	0.237	0.251	0.121
Lung Opacity	0.322	0.286	0.232	0.094
Pulmonary Fibrosis	0.257	0.288	0.196	0.068
Nodule/Mass	0.376	0.113	0.119	0.061
Consolidation	0.318	0.194	0.147	0.048
Pleural Thickening	0.233	0.228	0.151	0.050
Other Lesion	0.103	0.088	0.041	0.013
Calcification	0.095	0.023	0.026	0.012
Pneumothorax	0.022	0.111	0.011	0.005
All (Average)	0.304	0.275	0.241	0.121

4.3 Model Comparison — YOLOv8x vs YOLOv8s

Metric	YOLOv8x	YOLOv8s
Precision	~0.44	0.304
Recall	~0.03	0.275
mAP@50	~0.007	0.241
mAP@50-95	~0.002	0.121
Training Time	~8+ hours	~2.5 hours
Batch Size	4	2
Model Size	137 MB	22 MB
Best Performance Class	—	Cardiomegaly (0.672)

Key Finding: Despite being far smaller, **YOLOv8s significantly outperformed YOLOv8x** on this dataset. This is because YOLOv8x requires far more epochs and data to properly converge, while YOLOv8s converged efficiently within the hardware budget.

4.4 Analysis

Although YOLOv8x is architecturally superior, it underperformed due to: 1. **Insufficient epochs** — The extra-large model needed many more epochs to converge on 14-class medical data. 2. **Batch size constraint** — VRAM limitations forced a small batch size of 4, which destabilized the larger model’s learning. 3. **Dataset complexity** — Medical chest X-rays require domain-specific pre-training or much longer training runs for large models.

YOLOv8s, being a leaner model, converged more reliably within the same compute budget and achieved meaningfully better detection accuracy.

5. Prediction and Visualization

Both models were tested on the **260-image test set** using the YOLOv8 predict mode. Results were saved with bounding boxes drawn on each image.

Observations from YOLOv8s Predictions

- **Accurate detections:** Large structural abnormalities like Aortic Enlargement and Cardiomegaly were consistently and accurately detected.
- **Multiple detections per image:** Many images yielded 2–3 simultaneous correct detections across different disease classes.
- **No detections:** A portion of images (mostly faint lesions or healthy images) returned no bounding boxes — this is expected behaviour.
- **False positives:** Some anatomical features (ribs, vessels) were occasionally misidentified.

Visual comparison images (original raw X-ray vs model prediction) were generated and saved for all 260 test images at:

runs/detect/original_vs_predicted/

6. Error Analysis

The prediction results revealed several systematic limitations:

1. Missed Detections

- The model failed to detect small or subtle lesions (e.g., Calcification, Pneumothorax).
- Low contrast regions where pathology blends with background tissue were frequently missed.
- **Root Cause:** Insufficient representation of rare classes in training data.

2. False Positives

- Normal anatomical structures (vessel shadows, rib edges) were occasionally flagged as abnormalities.
- **Root Cause:** High visual similarity between certain normal structures and disease manifestations.

3. Low Recall on Rare Classes

- Calcification (recall: 0.023), Pneumothorax (recall: 0.111), and Other Lesion (recall: 0.088) had critically low recall.
- **Root Cause:** Class imbalance — these conditions have far fewer samples in the dataset.

4. Dataset Challenges

- Variability in DICOM-to-PNG conversion quality affected brightness and contrast.
 - Expert annotations had occasional inconsistencies in bounding box size.
 - Overlapping pathologies in a single image increased detection difficulty.
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7. Key Insights

- **Model size does not guarantee performance** — YOLOv8s outperformed YOLOv8x within the same training budget.
- **Compute budget matters** — Larger models require more epochs, larger batches, and more VRAM to reach their potential.

- **Class imbalance is the biggest challenge** — Rare conditions like Pneumothorax and Calcification are severely underdetected.
 - **Large structural findings are reliably detected** — Cardiomegaly (67.2% mAP) and Aortic Enlargement (67.0% mAP) demonstrate the model’s clear clinical utility.
 - **Domain-specific fine-tuning** on a pre-trained medical model would significantly improve all metrics.
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8. Conclusion

In Milestone 3, two YOLOv8 object detection models (YOLOv8x and YOLOv8s) were successfully trained, evaluated, and compared on the VinBigData Chest X-ray dataset containing 14 disease classes.

The **YOLOv8s model achieved superior performance** within the constraints of consumer-grade GPU hardware, reaching a mAP@50 of 0.241 overall, with individual class mAP@50 peaks of 0.672 for Cardiomegaly and 0.670 for Aortic Enlargement.

These results demonstrate the clinical potential of lightweight YOLO-based object detection for medical imaging, while also highlighting that dataset quality, class distribution balance, and hardware-appropriate model selection are equally important as architectural depth.

9. Future Work

- Apply **weighted loss functions** or **oversampling** to address class imbalance for rare diseases.
 - Use **cloud GPU** (e.g., Google Colab Pro, Kaggle T4) to train YOLOv8x for 100+ epochs with larger batches.
 - Fine-tune on a medical-specific pretrained backbone (e.g., a YOLO model pre-trained on chest X-ray data).
 - Implement **test-time augmentation (TTA)** to improve recall on subtle findings.
 - Build a real-time clinical detection interface using the trained YOLOv8s model.
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10. Summary

This milestone successfully demonstrates:

Achievement	Status
End-to-end model training (YOLOv8x & YOLOv8s)	Done
Performance evaluation with per-class metrics	Done
Comparative model analysis	Done
Test set prediction with bounding box visualization	Done
Original vs. Predicted image comparison saved	Done
Practical understanding of deep learning limitations	Done

The work establishes a strong, well-analyzed baseline for real-world medical chest X-ray abnormality detection.